Problem 1 Particle is loosing energy at a rate

$$\overline{P} = \frac{\mu_0 c q^2 \omega_B^2 \beta^2}{6\pi (1-\beta^2)^2}$$

Since $\omega_B = \frac{qB}{\gamma m} = \frac{\omega_0}{\gamma}$, $\gamma = \frac{1}{\sqrt{1-\beta^2}} \Rightarrow 1-\beta^2 = \frac{1}{\gamma^2}$

$$\beta^2 = 1 - \frac{1}{\gamma^2} = \frac{\gamma^2-1}{\gamma^2}$$

It is convenient to write everything in terms of γ: $\overline{P} = \frac{\mu_0 c q^2 \omega_0^2}{6\pi} \gamma^4 \frac{\beta^2}{\gamma^2} = A\gamma^2\beta^2$

$$A = \frac{\mu_0 c q^2 \omega_0^2}{6\pi}.$$

Finally $\overline{P} = A(\gamma^2-1)$

Thus $\frac{d}{dt}\gamma mc^2 = -A(\gamma^2-1)$

$$\frac{d\gamma}{\gamma^2-1} = -\frac{A}{mc^2}dt$$

$$\int_{\gamma_0}^{\gamma_t} d\gamma \frac{1}{\gamma^2-1} = \frac{1}{2}\int d\gamma \left(\frac{1}{\gamma-1} - \frac{1}{\gamma+1}\right) = \frac{1}{2}\ln\frac{\gamma-1}{\gamma+1}\Big|_{\gamma_0}^{\gamma_t}$$

$$\Rightarrow \ln\frac{\gamma_t-1}{\gamma_t+1} - \ln\frac{\gamma_0-1}{\gamma_0+1} = -\frac{2At}{mc^2}$$

$$\ln\left(\frac{\gamma_t+1}{\gamma_t-1}\cdot\frac{\gamma_0-1}{\gamma_0+1}\right) = \frac{2At}{mc^2} \equiv \alpha t.$$

$$\frac{\gamma_t+1}{\gamma_t-1} = \frac{\gamma_0+1}{\gamma_0-1} e^{\alpha t} \Rightarrow \gamma_t = \frac{\frac{\gamma_0+1}{\gamma_0-1} e^{\alpha t}+1}{\frac{\gamma_0+1}{\gamma_0-1} e^{\alpha t}-1}$$

or, $$\gamma_t = \frac{(\gamma_0+1)e^{\alpha t} + \gamma_0 - 1}{(\gamma_0+1)e^{\alpha t} - (\gamma_0-1)}$$

$$\mathcal{E}(t) = \gamma_t mc^2 \quad , \quad \mathcal{E}_0 = \gamma_0 mc^2$$

$$\mathcal{E}(t) = mc^2 \frac{(\mathcal{E}_0+mc^2)e^{\alpha t} + \mathcal{E}_0 - mc^2}{(\mathcal{E}_0+mc^2)e^{\alpha t} - (\mathcal{E}_0-mc^2)}$$

$$\alpha = \frac{2A}{mc^2} = \frac{\mu_0 c q^2 \omega_0^2}{3\pi mc^2}.$$

Note that $\mathcal{E}(t) \xrightarrow[t\to\infty]{} mc^2$, so that the particle is slowing down.

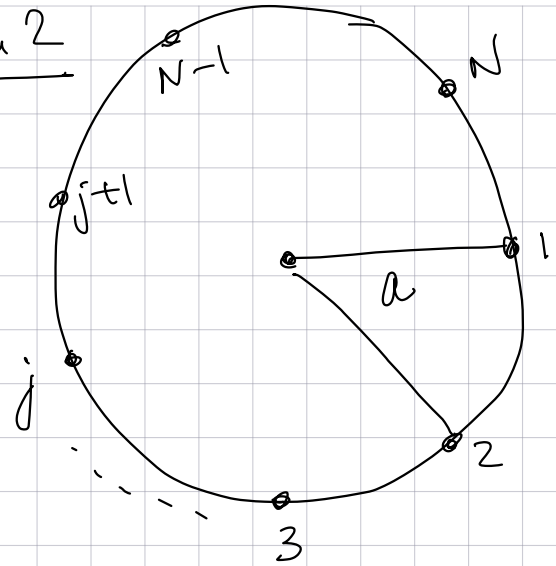
Radius of the circular trajectory changes as

$$a(t) = \frac{v(t)}{\omega_B} = \frac{v}{qB/\gamma m} = \frac{mc}{qB}(\beta\gamma) = \frac{mc}{qB}\sqrt{\gamma^2-1}$$

$$\Rightarrow a(t) = \frac{mc}{qB}\sqrt{\left(\frac{\mathcal{E}(t)}{mc^2}\right)^2 - 1} \longrightarrow 0 \text{ for } t\to\infty$$

because $\mathcal{E}(t) \to mc^2$.

Problem 2



N electrons on a circle of radius $a = \frac{v}{\omega}$

$$\vec{A}(t) = \sum_n \vec{A}_n e^{-in\omega_B t}$$

$$\vec{A}_n = \frac{\mu_0 q}{4\pi R T} \int \frac{dt\, e^{in\omega_B t}}{1 - \frac{\hat{n}\cdot\vec{v}(\tilde{t})}{c}} \vec{v}(\tilde{t})$$

where $\tilde{t}$ = retarded time, t = present time

$$t = \tilde{t} + \frac{1}{c}\left|\vec{R} - \vec{r}(\tilde{t})\right|, \quad dt = \left(1 - \frac{\hat{n}\cdot\vec{v}(\tilde{t})}{c}\right) d\tilde{t}$$

$$\Rightarrow \vec{A}_n = \frac{\mu_0 q}{4\pi R T} \int_0^T d\tilde{t}\; \vec{v}(\tilde{t})\, e^{in\omega\tilde{t} + in\frac{\omega}{c}\left(R - \hat{n}\cdot\vec{r}(\tilde{t})\right)}$$

where we used $|\vec{R} - \vec{r}| \simeq R - \hat{n}\cdot\vec{r}$.

Position of a given particle $\vec{r} = \frac{v}{\omega}\left(\sin(\omega\tilde{t} + \psi), -\cos(\omega\tilde{t} + \psi), 0\right)$

so that $\vec{v}(\tilde{t}) = v\left(\cos(\omega\tilde{t} + \psi), \sin(\omega\tilde{t} + \psi), 0\right)$

Here ψ is the phase shift of the particle which describes initial position of the particle on the circle.

Then $\hat{n}\cdot\vec{r} = \frac{v}{\omega} n_\perp \sin(\omega\tilde{t} + \psi) \Rightarrow \frac{n\omega}{c}\hat{n}\cdot\vec{r} = n\beta n_\perp \sin(\omega\tilde{t} + \psi)$

since $\vec{n} = (n_\perp, 0, n_z)$

So $A_{nx} = \frac{\mu_0 q v}{4\pi R T} e^{i k_n R} \int_0^T d\tilde{t} \cos(\omega \tilde{t} + \psi) e^{i n(\omega \tilde{t} + \psi - \psi)}$

$\cdot e^{-i n \beta n_\perp \sin(\omega \tilde{t} + \psi)}$

Let now $\varphi = \omega \tilde{t} + \psi \Rightarrow \frac{d\tilde{t}}{T} = \frac{d\varphi}{2\pi}$

so $\vec{A}_n = \frac{\mu_0 q v}{4\pi R} e^{i k_n R} \int_\psi^{2\pi+\psi} \frac{d\varphi}{2\pi} \begin{Bmatrix} \cos\varphi \\ \sin\varphi \end{Bmatrix} e^{i n \varphi - i n \beta n_\perp \sin\varphi}$

$\cdot e^{-i n \psi}$

But we have an identity $e^{-i n \beta n_\perp \sin\varphi} = \sum_{l=-\infty}^{\infty} J_l(n\beta n_\perp) e^{-i l \varphi}$

$$\vec{A}_n = \frac{\mu_0 q v}{4\pi R} e^{i k_n R} \int_\psi^{2\pi+\psi} \frac{d\varphi}{2\pi} \sum_l J_l(n\beta n_\perp) \begin{Bmatrix} \cos\varphi \\ \sin\varphi \end{Bmatrix} e^{i n \varphi - i l \varphi}$$

The φ-integral is now easy: $\int_\psi^{2\pi+\psi} \frac{d\varphi}{2\pi} e^{i(m-l)\varphi} =$

$$= \frac{e^{i(m-l)(2\pi+\psi)} - e^{i(m-l)\psi}}{i(m-l)} = e^{i(m-l)\psi} \frac{e^{i 2\pi (m-l)} - 1}{i(m-l)} =$$

$= 0$ unless $\underline{m = l}$ (because $m - l =$ Integer)

That is, $\int_\psi^{2\pi+\psi} d\varphi\, e^{i(m-l)\varphi} = 2\pi \delta_{m,l}$ where $m = n \pm 1$.

Then $A_{nx} \sim \frac{1}{2}\left(J_{n+1}(n\beta n_\perp) + J_{n-1}(n\beta n_\perp)\right)$

while $A_{ny} \sim \frac{1}{2i}\left(J_{n+1}(n\beta n_\perp) - J_{n-1}(n\beta n_\perp)\right)$.

Hence $\vec{A}_m = \frac{\mu_0 q v}{4\pi R} e^{i k_m R} \cdot \left(e^{-i m \psi}\right)$

$\cdot \begin{cases} J_m(m\beta n_\perp)/\beta n_\perp & \text{for } x\text{-component} \\ i J_m'(m\beta n_\perp) & \text{for } y\text{-component} \end{cases}$

just like in the lecture except for $e^{-i n \psi}$

Hence for N-particles with phase shifts ψ_j

we will have

$$\vec{A}_m = \frac{\mu_0 q v}{4\pi R} e^{i k_m R} \left(\frac{J_m(\beta m n_\perp)}{\beta n_\perp}, i J_m'(\beta m n_\perp), 0 \right) \cdot$$

$$\cdot \sum_{j=1}^{N} e^{-im\psi_j} = \vec{A}_m^{(0)} \cdot \sum_{j=1}^{N} e^{-im\psi_j}$$

vector potential of 1 particle.

Then $\vec{A}(t) = \sum_m \vec{A}_m e^{-im\omega t}$

$$= \sum_m \sum_{j=1}^{N} \vec{A}_m^{(0)} e^{-im\omega t} e^{-im\psi_j}$$

sum over harmonics

sum over particles

Calculating intensity (averaged over the period T)

we get $\frac{dP}{d\Omega} = \hat{n} \frac{2cR^2}{\mu_0} \sum_{m=1}^{\infty} |\vec{B}_m|^2$, where $\vec{B}_m = i k_m \hat{n} \times \vec{A}_m$

But $\vec{B}_m = \vec{B}_m^{(0)} \cdot \sum_{j=1}^{N} e^{-im\psi_j}$

field of 1 particle

$$\Rightarrow |\vec{B}_m|^2 = |\vec{B}_m^{(0)}|^2 \left(\sum_{j=1}^{N} e^{im\psi_j} \right) \left(\sum_{j'=1}^{N} e^{-im\psi_{j'}} \right)$$

$$= |\vec{B}_m^{(0)}|^2 \sum_{j,j'=1}^{N} \cos m(\psi_j - \psi_{j'}) = |\vec{B}_m^{(0)}|^2 \cdot S_N^{(m)}$$

$$S_N^{(m)} = N + \sum_{j' \neq j}^{N} \cos m(\psi_j - \psi_{j'}).$$

$$S_N^{(m)} = N + \sum_{j' \neq j}^{N}{}' \cos m(\psi_j - \psi_{j'})$$ because we separated out $j = j'$ term (it gives N).

Thus $$\frac{dP}{d\Omega} = \sum_m I_m S_N^{(m)}$$

(i) Now, if $\{\psi_j\}$ are completely random than, for fixed j', $\sum_{j=1}^{N}{}' \cos m(\psi_j - \psi_{j'}) = 0$.

We obtain $\frac{dP}{d\Omega} = N \cdot \sum_m I_m$ so that intensity due to N particles is <u>incoherent</u> sum of individual intensities.

(ii) Let $\psi_j = \frac{2\pi}{N}(j-1)$, $j = 1, 2 \ldots N$

Write $S_N^{(m)}$ as unrestricted sum again

$$S_N^{(m)} = \sum_{j'=1}^{N} \sum_{j=1}^{N} \cos \frac{2\pi m}{N}\left(j-1-(j'-1)\right) = \frac{1}{2} \sum_{j'=1}^{N} \sum_{j=1}^{N} \left[e^{i\frac{2\pi m}{N} j} \quad e^{-i\frac{2\pi m}{N} j'} + c.c. \right] =$$

$$= \sum_{j=1}^{N} \left(e^{i\frac{2\pi m}{N}}\right)^j \sum_{j'=1}^{N} \left(e^{-i\frac{2\pi m}{N}}\right)^{j'}.$$

Let $s = \sum_{j=1}^{N} r^j = r + r^2 + \ldots + r^{N-1} + r^N = 1 + r + \ldots + r^{N-1}$.

$r = e^{i 2\pi m/N} \Rightarrow r^N = e^{i 2\pi m} = 1$

Now, $r \cdot s = r(1 + r + \dots + r^{N-1}) = r + r^2 + \dots +$

$$\underset{\substack{\shortparallel \\ 1}}{+ r^N} = 1 + r + \dots + r^{N-1} = s.$$

$\Rightarrow$ we get $rs = s$ and $r = e^{i2\pi m/N} \neq 1$.

$\Rightarrow s = 0$.

This is so unless $r = 1$ which happens for $m = pN$, where <u>p = integer</u>.

In this case $e^{i\frac{2\pi}{N} \cdot pN} = e^{i2\pi p} = 1$ and

$$\sum_{j=1}^{N} r^j = \sum_{j=1}^{N} 1 = N.$$

$\Rightarrow$ hence $S_N^{(m)} = \begin{cases} 0, & \text{for } m \neq pN \\ N^2, & \text{for } m = pN. \end{cases}$

$$\Rightarrow \frac{dP}{d\Omega} = \sum_{m=1}^{\infty} I_m S_N^{(m)} = N^2 \sum_{p=1}^{\infty} I_{m=pN}$$

We obtained <u>coherent</u> enhancement by N^2 of frequencies $m = p \cdot N$, where $p = 1, 2, 3 \dots$.

At the same time, all other frequencies with $m \neq pN$ are completely <u>suppressed</u>.